

Software Architecture Document

for

Draftly

*A Lawyer-in-the-Loop Legal Workflow Platform for Sri Lankan
Legal Practice*

Version 1.0 | Product Release v0

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Revision History

Version	Date	Description	Author
0.1	29 July 2026	Initial architecture for the V0 RTA workbench	Project team

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1. Introduction

1.1 Purpose

This document defines the software architecture for the initial release of a lawyer-controlled notarial and conveyancing workbench for Sri Lankan Registration of Title Act (RTA) matters. It translates the approved V0 requirements into components, interfaces, data structures, processing sequences, deployment decisions, and quality controls.

The intended readers are the project team, supervisors, legal domain reviewers, developers, testers, security reviewers, and future maintainers. The document is also the design baseline against which implementation and test evidence can be traced.

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1.2 Scope

V0 supports examination of title, drafting, execution preparation, and attestation preparation for an RTA matter. Its first active instrument flow is the approved transfer form. The architecture supports:

- secure matter and document management;
- Google Document AI processing for supported printed and clear scanned documents;
- source-linked candidate particulars and lawyer verification;
- a verified structured matter record;
- deterministic document, identity, consistency, form, execution, and attestation checks;
- grounded research over a controlled Sri Lankan legal-source corpus;
- guided legal tasks with recorded decisions and overrides;
- generation from approved templates and verified particulars;
- lawyer review, approval, versioning, DOCX/PDF export, and audit; and
- English and Sinhala interfaces and outputs for the supported path.

V0 does not automate registration, reconstruct a historical chain of title, extract Abstract of Title forms, accept unreliable handwriting or severely degraded scans as verified data, or provide legal advice without lawyer control. Unsupported documents remain available as evidence and are routed to manual review. Later releases may add harder-document processing, more RTA forms, wider conveyancing regimes, collaboration, integrations, and organisation-scale operation without bypassing the evidence, verification, approval, privacy, and audit controls established here.

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1.3 Definitions, Acronyms, and Abbreviations

Term	Meaning
API	Application Programming Interface
Candidate particular	A machine-extracted value awaiting lawyer review
Controlled content	Versioned legal rules, sources, terminology, task definitions, and templates governed by authorised users
Document derivative	OCR text, layout, preview, thumbnail, or extraction output created from an original document
Evidence span	A resolvable page, region, or text range in a specific document version
Finding	The result of a deterministic check: pass, warning, fail, or needs review
OCR	Optical Character Recognition
OIDC	OpenID Connect
Original document	The immutable file submitted as matter evidence
PDPA	Personal Data Protection Act, No. 9 of 2022
RTA	Registration of Title Act, No. 21 of 1998
Verified matter record	Lawyer-verified and corrected particulars that are authoritative for generated matter facts
V0	The initial, RTA-first product release

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1.4 References

The formal external references appear in Section 12. This architecture is derived from the V0 Software Requirements Specification and the approved RTA scope, legal-source inventory, document-processing research, and interface research. Those project artefacts are inputs to the design, not runtime dependencies.

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1.5 Overview

Section 2 introduces the architectural views. Section 3 records goals, constraints, and decisions. Sections 4 to 9 present the use-case, logical, process, deployment, implementation, and data views. Sections 10 and 11 cover capacity, performance, and quality. Section 12 lists the cited sources

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2. Architectural Representation

The architecture uses six complementary views. Each view describes the same system boundary from a different concern and shares the contracts defined in the data view.

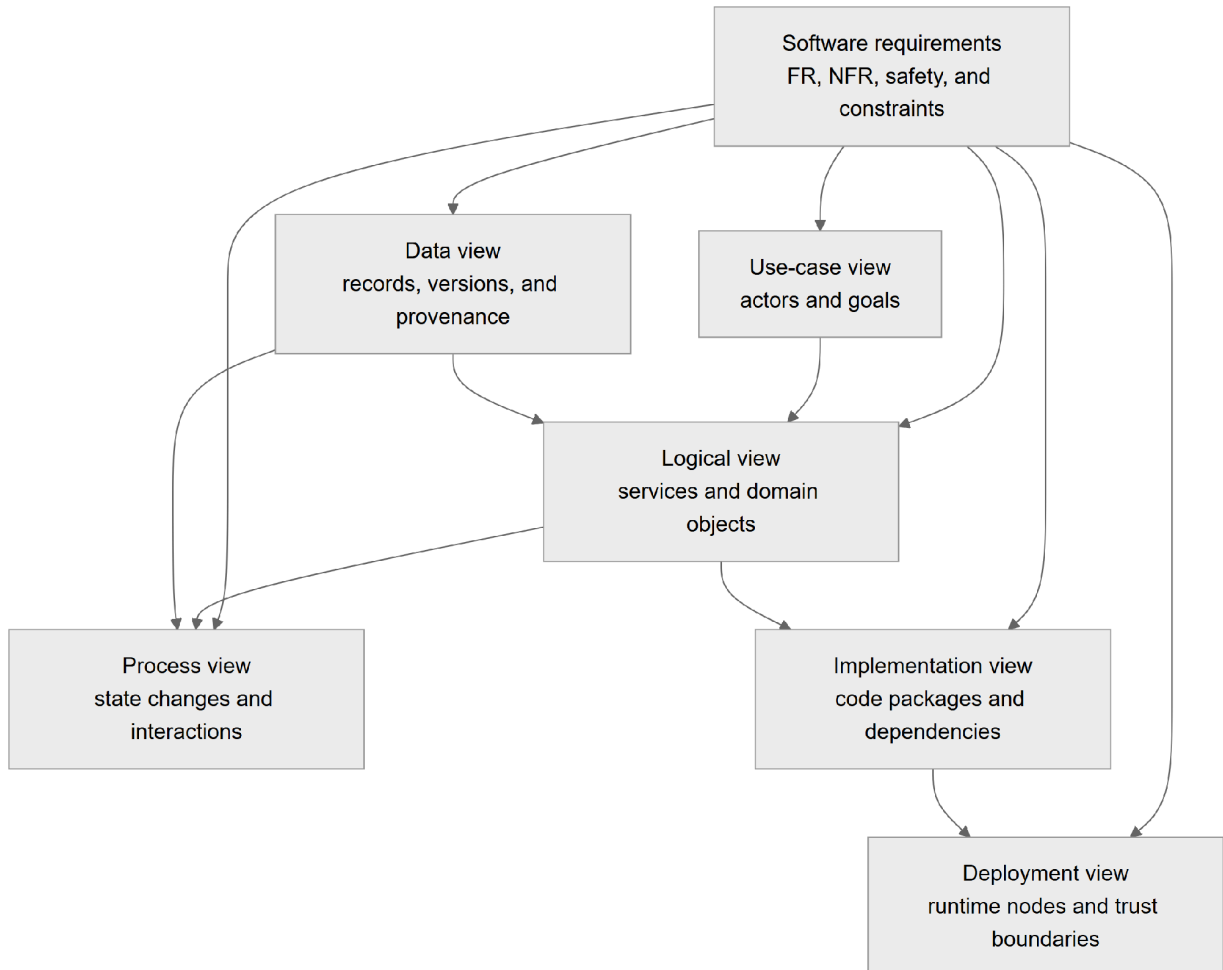


Figure 1. Architectural views and their relationships.

Figure 1 makes traceability explicit. A requirement is realised by one or more use cases, logical components, processes, deployment nodes, implementation packages, and data records. Tests can therefore trace a requirement to both the responsible component and the persistent evidence of its execution.

The complete document landscape is shown separately because a legal workbench must distinguish evidence, machine derivatives, controlled authority, internal records, and approved work products.

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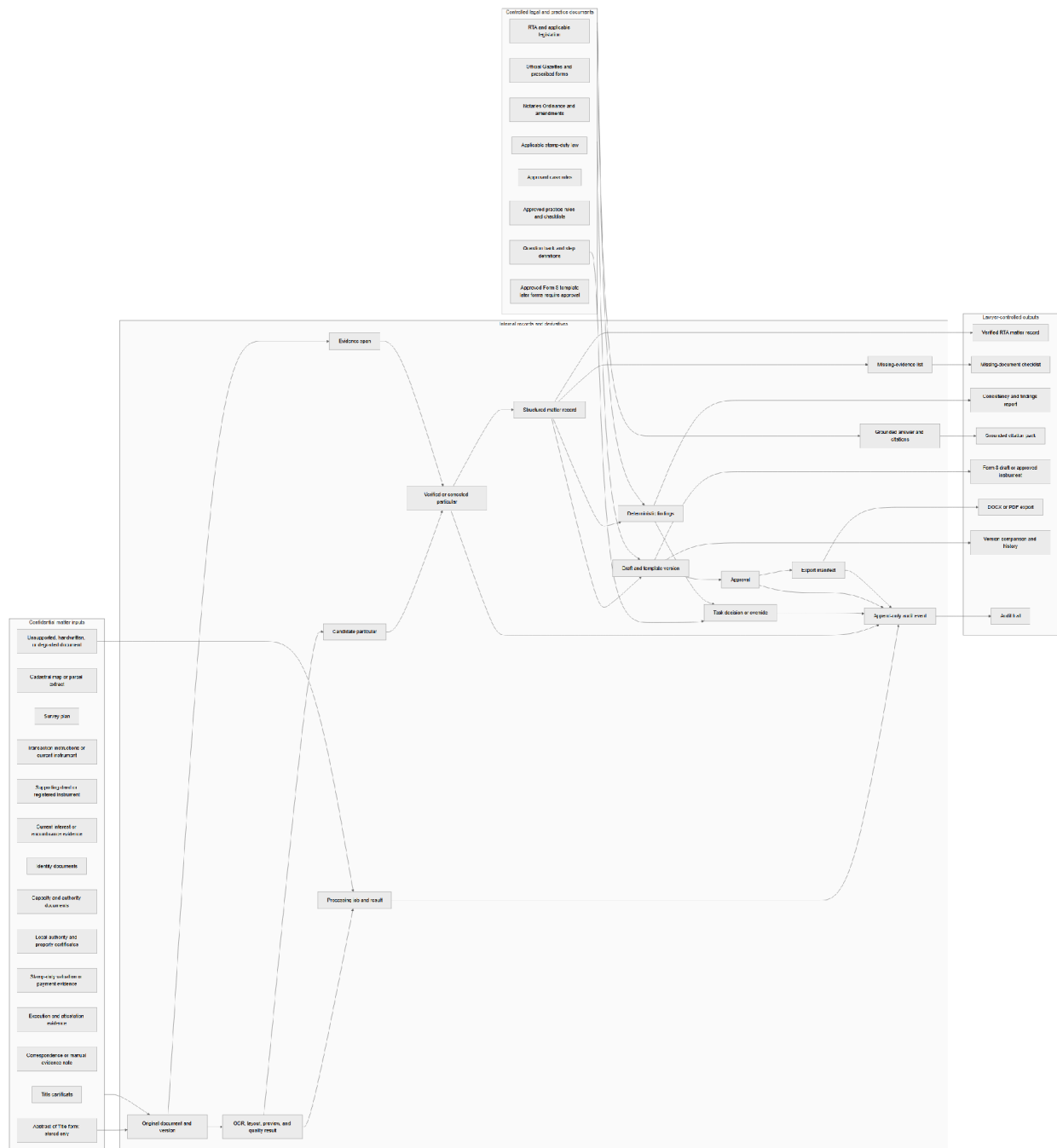


Figure 2. Complete V0 document and record ecosystem.

Figure 2 includes every document family that can enter, support, be created within, or leave the V0 system. An arrow indicates controlled use or derivation, not automatic legal acceptance. In particular, OCR output becomes a candidate particular; only lawyer verification can place a value in the verified matter record.

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2.1 Document Inventory and Treatment

Group	Documents or records	Architectural treatment
Title and parcel	Current title certificate, cadastral map or parcel extract, survey plan	Preserved originals; approved fields extracted; identifiers cross-checked
Transaction	Instructions, proposed/current instrument, supporting deed or current registered instrument	Evidence for the present transaction; not used to infer an unsupported historical chain
Interests	Mortgage, lease, life interest, caveat, injunction, and other current registered-interest evidence	Structured as current interests with source spans and review states
Identity and capacity	NIC/passport or approved identity evidence; power of attorney; corporate authority; death, probate, administration, or estate authority	Access-restricted; minimum display; lawyer verification required
Local authority and property	Assessment notice, rates receipt or clearance, street-line, building-line, non-vesting, ownership, conformity, and building-approval records	Checked according to approved applicability rules; no checklist item becomes law without authority
Financial and execution	Stamp-duty valuation/payment evidence; witness details; signature, photograph, alteration, erasure, and attestation evidence	Used by deterministic drafting, execution, and attestation checks
Supporting material	Correspondence, manual evidence notes, photos, and other supporting files	Stored and classified; manual claims identify author, reason, and source status
Unsupported V0 input	Handwriting, severely degraded scans, unsupported scripts, and Abstract of Title forms	Original preserved; explicit quality/unsupported state; manual review or entry; no automatic acceptance
Primary legal authority	RTA, official Gazettes, Notaries legislation, and applicable stamp-duty law	Separate versioned corpus; segmented with provenance, effective date, and authority level
Controlled guidance	Approved case rules, practice rules, checklists, terminology, question bank, step definitions,	Versioned, reviewed, activated, retired, and audited; derived guidance links to approved

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Group	Documents or records	Architectural treatment
	and templates	authority
Processing records	Document version, checksum, preview, OCR/layout, quality result, processing job, processor/version/region, and evidence span	Rebuildable derivatives remain separate from immutable originals and authoritative verified data
Matter records	Candidate, verified, corrected, conflicting, blocked, and manual particulars; parties, parcels, instruments, interests, findings, task decisions	Versioned domain records with provenance and explicit state transitions
Drafting records	Template version, draft snapshot, version hash, approval, export manifest	Draft generation accepts verified/corrected particulars only; post-approval edits create a new unapproved version
User outputs	Verified matter record, missing-evidence checklist, findings report, citation pack, Form 8 draft, DOCX/PDF, comparison, and audit trail	Permission-controlled views or generated artefacts, each identifying source and version

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3. Architectural Goals and Constraints

3.1 Goals

The architecture has seven priorities, in order:

1. Prevent unverified matter facts and unsupported legal claims from entering an approved output.
2. Preserve every original document and make every accepted extracted particular traceable to its exact version and source location.
3. Keep confidential matter information isolated by user, matter, storage boundary, and query.
4. Make corrections, decisions, overrides, content changes, approvals, and exports reconstructable from append-only history.
5. Provide recoverable failure and explicit abstention instead of silent loss or fabricated completion.
6. Meet the V0 response, processing, bilingual, accessibility, and export targets.
7. Allow later forms and conveyancing regimes to extend common contracts without weakening V0 controls.

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3.2 Constraints

- The product is browser-based and applies authorisation on the server.
- Google Document AI is the V0 OCR and document-quality provider. A provider adapter records processor, version, region, purpose, timing, and outcome.
- Original evidence and derivatives use separate storage namespaces.
- Confidential matter records and the public or controlled legal corpus use separate repositories and access paths.
- The verified matter record is the only authoritative source of generated matter facts.
- Deterministic checks do not depend on generated prose.
- Ordinary application users cannot edit or delete audit history.
- The core V0 path supports English and Sinhala display and export.
- No copied third-party interface, text, logo, or proprietary asset forms part of the product.
- Provider, retention, deployment-region, and professional-conduct approvals must be completed before live client use.

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3.3 Principal Architecture Decisions

Decision	Selected V0 approach	Reason and consequence
System shape	Modular monolith API plus separate asynchronous workers	Keeps transactions and policy enforcement coherent while isolating long-running OCR, retrieval, and export work
Web client	Next.js with React and TypeScript	Supports a typed, accessible browser workbench and clear route/package boundaries
API	FastAPI with versioned OpenAPI and Pydantic contracts	Matches the Python document/retrieval work and validates every boundary
Matter database	PostgreSQL	Supports transactions, JSON where needed, indexes, migrations, and defence-in-depth row security
Original storage	Protected object storage with immutable version references	Large evidence files do not become public paths or database blobs
Background work	Authenticated queue with idempotent jobs and retry policy	Upload remains responsive; provider failures become visible and recoverable
Document processing	Provider-neutral port with a Google Document AI adapter	Satisfies the selected V0 service while preserving an exit and manual path
Legal retrieval	Separate, versioned legal corpus and derived search index	Prevents accidental retrieval of confidential matter documents and preserves authority metadata
Checks	Versioned deterministic rule catalogue	Results are repeatable, testable, and legally reviewable
Draft editor	Schema-validated Tiptap JSON with protected wording and fact references	Makes template structure, provenance, and version comparison enforceable
Export	Server-side DOCX/PDF generation from an approved snapshot	Produces a stable artefact independent of browser rendering and records an export manifest
Authentication	OIDC-compatible identity provider and server-side role/matter checks	Avoids local password design and centralises identity lifecycle while retaining application authorisation

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The exact managed-service products, deployment region, Document AI processor version, retention periods, and production identity provider remain approval items. Their interfaces and required controls are fixed here so a selection does not alter domain behaviour.

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4. Use-Case View

4.1 Actors and Use Cases

Actor	Responsibilities
Lawyer	Owns legal verification, findings decisions, final approval, and professional judgement
Legal clerk	Prepares matters, uploads documents, reviews candidates, and performs permitted corrections without final approval
Legal-content administrator	Curates source mappings, rules, task definitions, terminology, and templates
System administrator	Manages accounts, roles, configuration, monitoring, and recovery without acquiring unrestricted matter access
External identity service	Authenticates users and returns trusted identity claims
Google Document AI	Returns OCR, layout, and quality output for an authorised processing request

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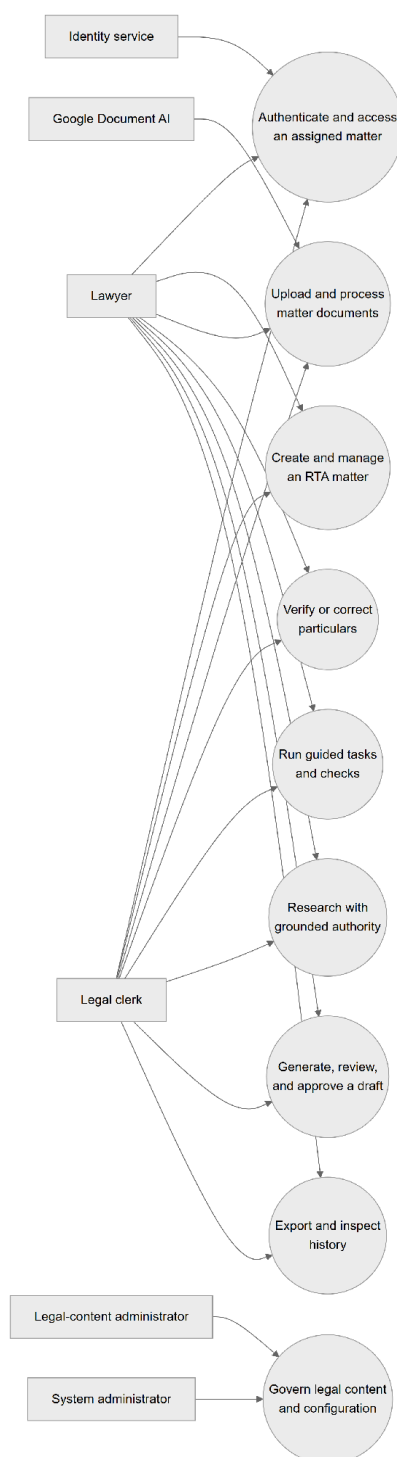


Figure 3. VO use-case diagram.

Figure 3 separates preparation from legal authority. A clerk may prepare and correct data where permitted, but only a lawyer may assign final verification, resolve or override legal findings where authorised, and approve a work product. External services participate through

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narrow interfaces and do not become actors with approval power.

4.2 Use-Case Realisations

4.2.1 UC-02: Create and Manage an RTA Matter

Field	Realisation
Actor	Lawyer or legal clerk
Description	Creates a privacy-safe RTA matter and assigns permitted users
Preconditions	Authenticated user has matter-creation permission
Main flow	Enter matter reference and transaction type; select RTA regime; assign users; save; open matter overview
Success condition	Matter and membership records exist and an audit event records creation
Fail condition	Invalid data or unauthorised assignment is rejected without partial creation
Extensions	Archive/reopen matter; change assignment; record reason for restricted action

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4.2.2 UC-03: Upload and Process Matter Documents

Field	Realisation
Actor	Lawyer or legal clerk; Google Document AI
Description	Preserves an original and produces reviewable OCR and candidate particulars
Preconditions	User can access the matter; file type and size pass validation
Main flow	Upload; checksum and store original; create document version; enqueue processing; classify; call Document AI; store derivatives and candidates; show review state
Success condition	Every accepted file reaches ready-for-review or an explicit failure/manual-review state
Fail condition	Original is preserved and a recoverable failure is displayed; no candidate is verified automatically
Extensions	Duplicate warning; unsupported handwriting/degraded scan; retry; manual classification; replacement creates a new version

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4.2.3 UC-04: Verify or Correct Particulars

Field	Realisation
Actor	Lawyer; legal clerk within permitted preparation scope
Description	Compares a candidate or manual value with source evidence and records a decision
Preconditions	User can access the matter and evidence version
Main flow	Open candidate; view source page/region; compare conflicts; verify, correct, reject, or block; record reviewer and reason
Success condition	A new particular version and audit event preserve the decision and provenance
Fail condition	Missing evidence, lost permission, or stale version prevents the transition
Extensions	Manual entry with evidence note; batch review only after evidence is displayed; supersede an earlier value

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4.2.4 UC-05: Run Guided Tasks and Checks

Field	Realisation
Actor	Lawyer or legal clerk
Description	Runs approved, applicable task definitions and deterministic checks over verified data
Preconditions	Matter exists; active rule and task versions are selected
Main flow	Determine applicability; identify missing inputs; execute checks; create findings; inspect authority and evidence; record decision
Success condition	Step run records inputs, rule versions, results, and authorised decisions
Fail condition	Missing evidence yields needs-review/blocked, not a false pass
Extensions	Supply evidence; rerun after correction; lawyer resolves or records a reasoned override

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4.2.5 UC-06: Research with Grounded Authority

Field	Realisation
Actor	Lawyer or legal clerk
Description	Searches the controlled corpus and produces claim-level citations
Preconditions	User is authenticated; legal-corpus version is available
Main flow	Submit query and optional matter scope; retrieve sources; rank passages; compose bounded claims; validate citations; display answer and exact passages
Success condition	Every retained claim resolves to supporting authority with status and effective-date metadata
Fail condition	Unsupported parts explicitly abstain; no invented citation is shown
Extensions	Search without generation; filter by source type; mark a candidate case rule for legal review

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4.2.6 UC-07: Generate, Review, and Approve a Draft

Field	Realisation
Actor	Lawyer; legal clerk for preparation
Description	Creates a Form 8 draft from an approved template and verified particulars
Preconditions	Active approved template exists; mandatory facts are verified/corrected; blocking findings are resolved or validly overridden
Main flow	Select template; bind verified facts; validate locked wording and placeholders; save snapshot; review evidence; lawyer approves version
Success condition	Approval identifies approver, exact content hash, template version, fact versions, and time
Fail condition	Unverified mandatory fact, placeholder, changed locked text, or unresolved blocker prevents approval
Extensions	Edit creates a new unapproved version; compare/restore; request missing evidence

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4.2.7 UC-08: Export and Inspect History

Field	Realisation
Actor	Lawyer or authorised legal clerk
Description	Produces a controlled DOCX/PDF and reconstructs significant matter activity
Preconditions	User has export permission; selected artefact version is approved where approval is required
Main flow	Request export; generate from immutable snapshot; validate fonts/placeholders/version; store artefact and manifest; record audit event; provide expiring download
Success condition	Reopened output preserves English/Sinhala content and identifies the approved version
Fail condition	Generation or validation failure produces no misleading completed export
Extensions	Revoke download; re-export same snapshot; view correction, decision, approval, and permission history

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4.2.8 UC-09: Govern Legal Content

Field	Realisation
Actor	Legal-content administrator
Description	Versions and controls legal sources, rules, steps, terminology, and templates
Preconditions	User has content-governance role
Main flow	Create draft version; attach authority and effective date; run validation/tests; request legal approval; activate; retire or supersede
Success condition	Active content has owner, approval, effective period, version, tests, and audit event
Fail condition	Incomplete authority, failed tests, or unauthorised approval prevents activation
Extensions	Roll back by activating an earlier approved version; rebuild derived search index

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5. Logical View

5.1 Overview

The V0 API is a modular monolith: domain modules execute in one deployable API and one transactional database, but communicate through explicit service and repository interfaces. Document processing, retrieval/answer work, and export rendering execute in separate workers because they are long-running, provider-facing, or resource intensive.

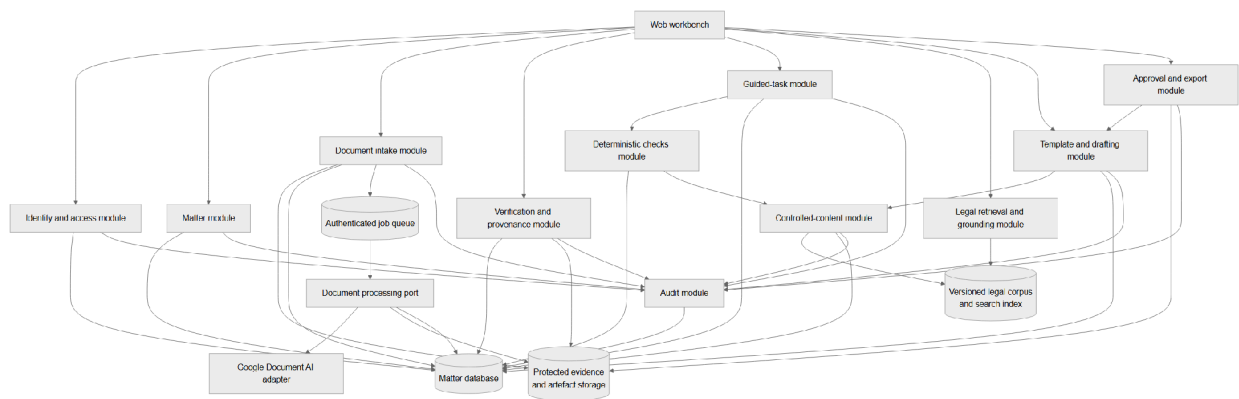


Figure 4. Logical components and repositories.

Figure 4 places policy at the domain boundary, not in the browser. Audit is a cross-cutting module called inside the same transaction as each significant state change. The legal corpus is read through the retrieval/content modules; matter modules cannot search confidential evidence as if it were public law.

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5.2 Architecturally Significant Design Packages

Package	Responsibility	Owned records
Identity and access	Identity claims, roles, matter membership, action policy	User, Role, MatterMembership
Matter	Matter lifecycle and current structured entities	Matter, Party, Parcel, Instrument, Interest
Documents	Upload, validation, immutable versioning, classification, processing state	Document, DocumentVersion, ProcessingJob, ProcessingResult
Verification	Evidence viewer, candidate review, corrections, conflicts, provenance	SourceSpan, Particular, ParticularVersion
Guidance	Applicable tasks, step runs, decisions, blocks, overrides	StepDefinition reference, StepRun
Checks	Deterministic rule execution and findings	RuleDefinition reference, Finding
Retrieval	Corpus search, passage ranking, bounded answer, citation validation	SearchRun, GroundedAnswer, Citation
Drafting	Template binding, editor schema, snapshots, placeholder and wording validation	Template reference, Draft, DraftVersion
Approval and export	Approval gate, rendering, manifest, download revocation	Approval, Export, ExportManifest
Controlled content	Source, rule, step, terminology, and template governance	LegalSource, LegalSection, CaseRule, RuleDefinition, StepDefinition, Template
Audit	Append-only event creation and authorised history queries	AuditEvent

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5.3 Domain Class Model

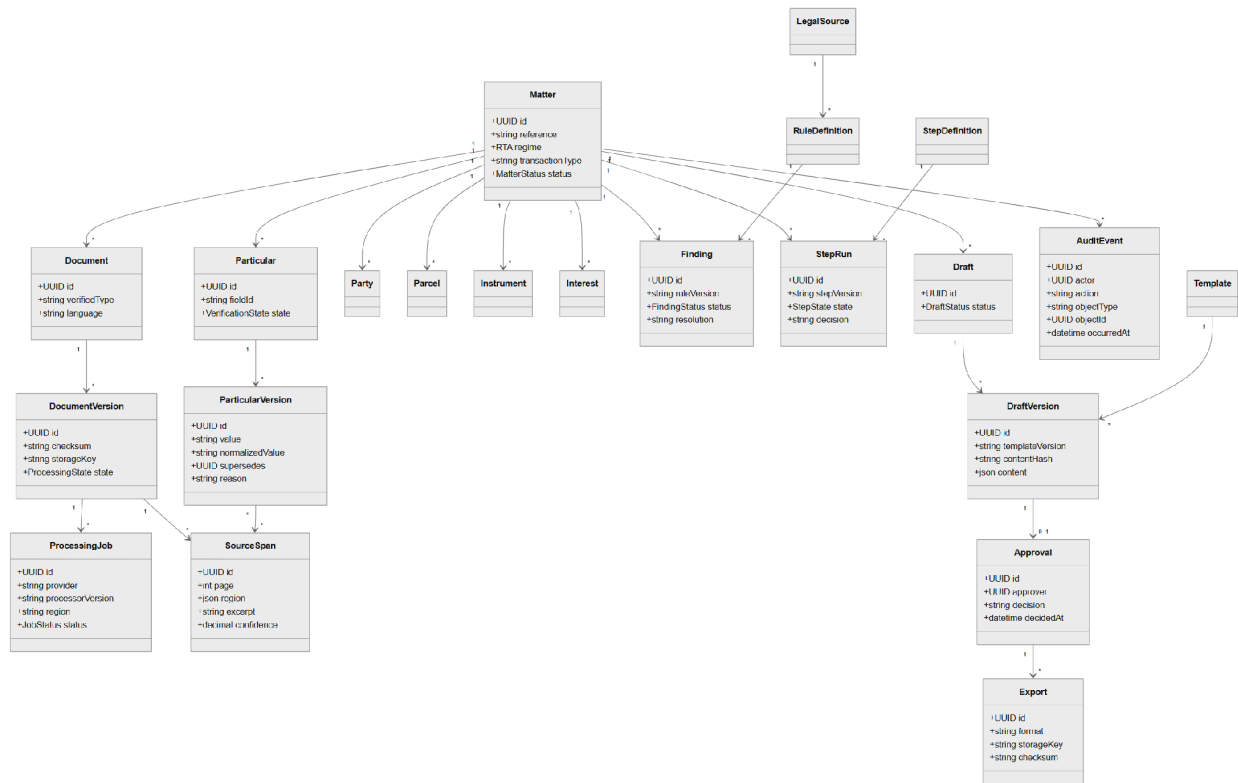


Figure 5. Significant domain classes and provenance relationships.

Figure 5 shows the records that enforce legal safety. A particular is not overwritten: each correction creates a new version, and its evidence links target an immutable document version. Draft approval targets a content hash, so a later edit cannot retain the old approval.

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6. Process View

6.1 Document-to-Draft Activity

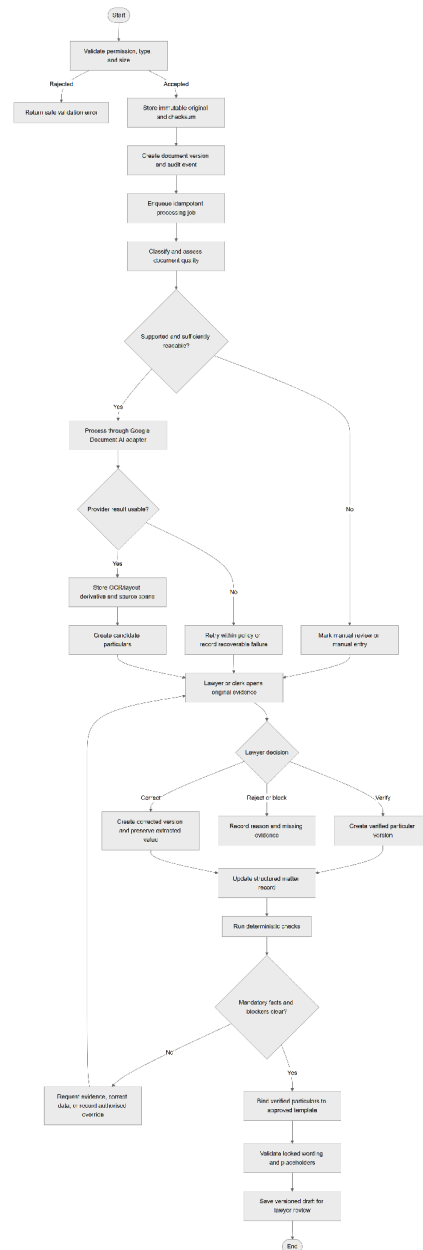


Figure 6. Document intake, verification, checking, and draft preparation.

The process never turns OCR confidence into legal verification. Every branch ends in a visible review, failure, or blocked state. The loop after checks is deliberate: the user resolves the underlying evidence or records a permitted lawyer decision rather than editing a report until it appears to pass.

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6.2 Document Processing Sequence

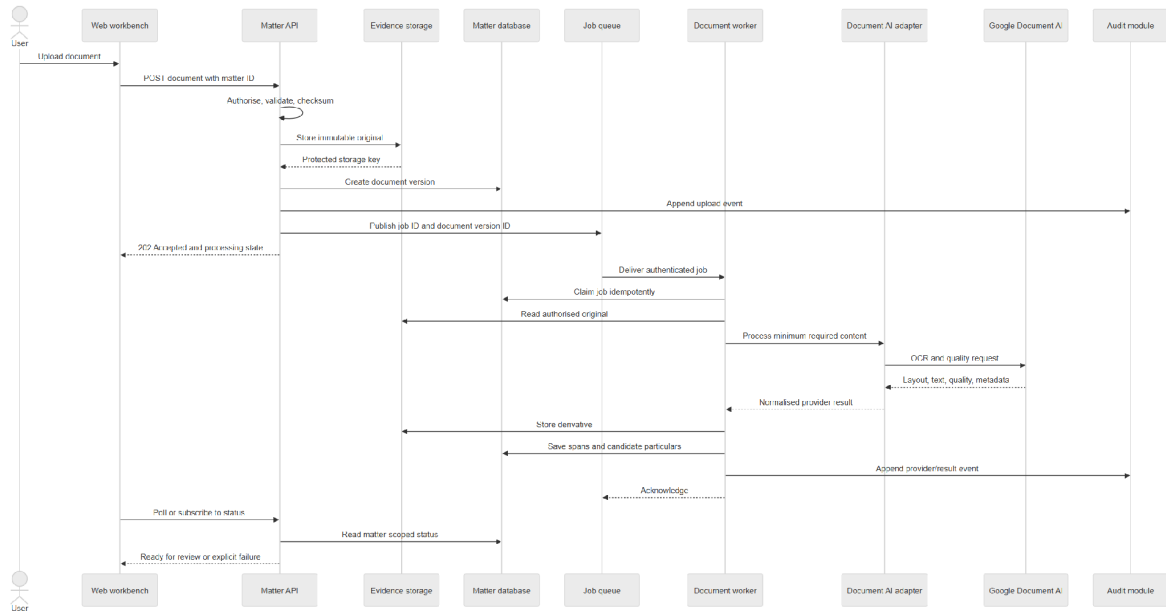


Figure 7. Internal sequence for asynchronous document processing.

Figure 7 shows the objects hidden by the simple user action. The API returns after durable acceptance; a worker owns provider latency and retry. The queue contains identifiers, not the complete confidential file, and every matter-scoped read is authorised again.

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6.3 Grounded Research Sequence

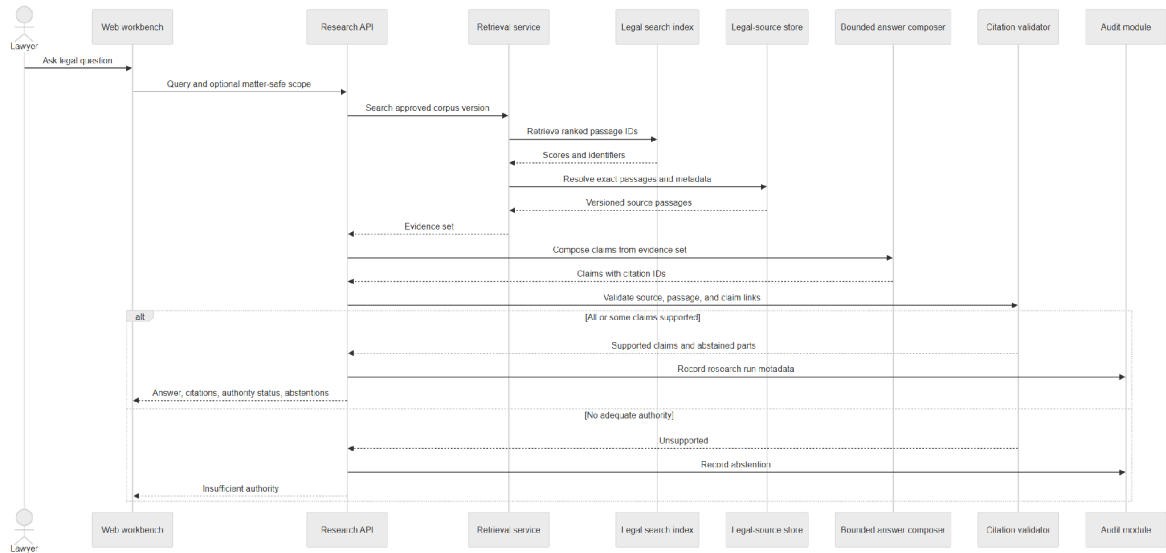


Figure 8. Grounded legal-research sequence.

The composer cannot write a citation into the corpus or activate a candidate rule. The validator resolves every citation against the versioned source store and permits partial or complete abstention. Matter content, when used to frame a query, is not inserted into the public legal corpus.

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6.4 Approval and Export Sequence

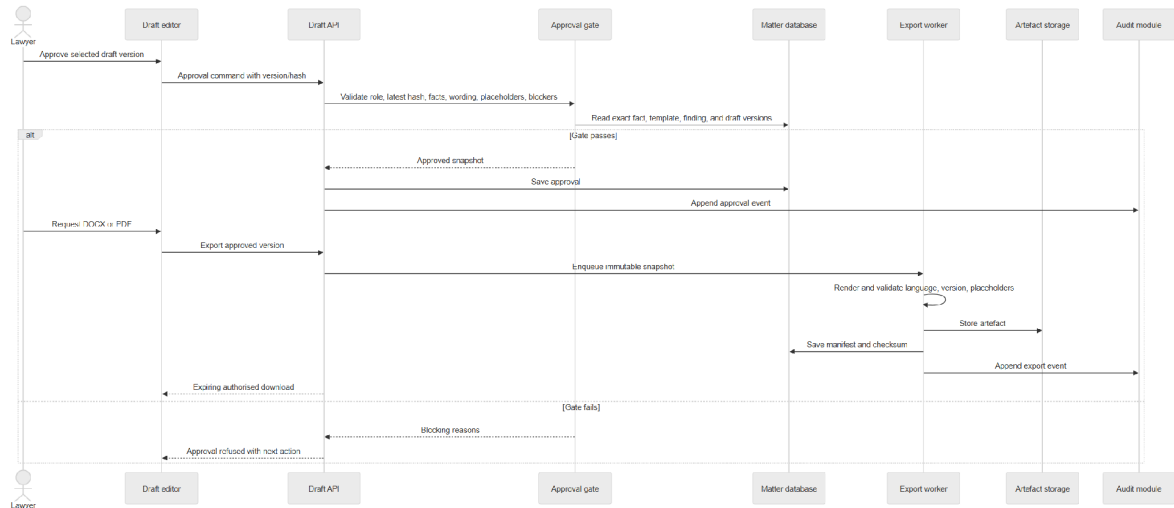


Figure 9. Approval and controlled-export sequence.

Approval is an atomic decision about one immutable draft version. Export renders that snapshot, not the mutable editor state. The manifest links the output checksum to the approval, template, fact versions, application version, and renderer version.

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7. Deployment View

7.1 Reference Deployment

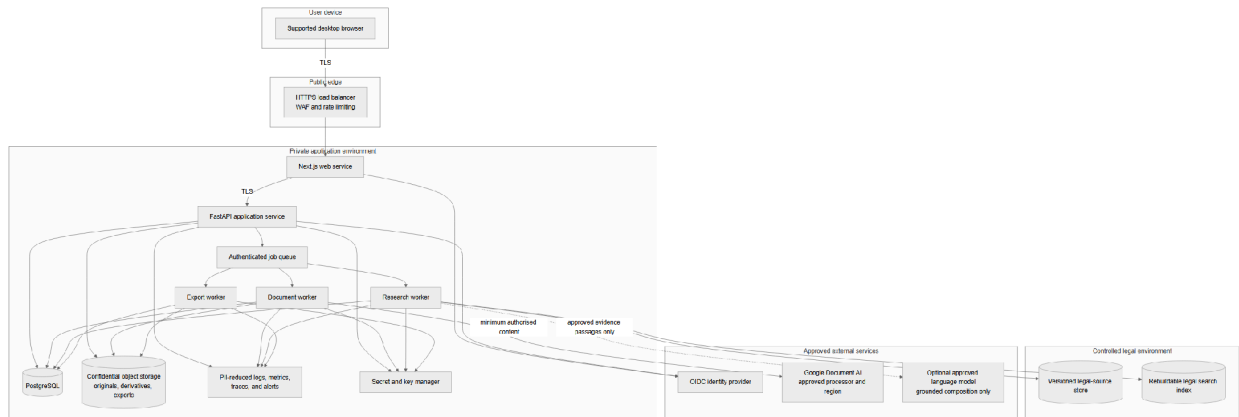


Figure 10. Reference V0 deployment and trust boundaries.

Figure 10 is cloud-provider neutral except for the selected Document AI dependency. The database and storage are not public. Workers have separate service identities and minimum permissions; for example, a research worker cannot read matter originals, and an export worker can read only the approved snapshot and required template assets.

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7.2 Deployment Controls

Boundary	Required controls
Browser to edge	TLS, secure cookies, CSRF protection where applicable, content security policy, safe file-upload feedback
Edge to services	Authenticated internal requests, request limits, correlation IDs, no trust in client-supplied role or matter ID
Service to database	Least-privilege database roles, transactional audit writes, encrypted connections, migration control, backup and restore tests
Service to storage	Opaque keys, short-lived signed access, encryption, malware/type validation, original/derivative/export separation
Queue and workers	Authenticated publisher/consumer, idempotency key, lease/visibility timeout, bounded retry, dead-letter/manual-review outcome
External processor	Approved region/version, minimum content, restricted credentials, timeout, retention/training controls, provider audit metadata
Legal corpus	Read-only serving version, controlled ingestion, effective-date and authority metadata, rebuildable index
Observability	No raw identity numbers, signatures, full addresses, pages, or generated legal content in ordinary logs

Production network topology, recovery objectives, retention policy, and region must be approved before live deployment. Development and demonstration deployments use synthetic or approved anonymised documents only.

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8. Implementation View

8.1 Overview

The implementation keeps framework code at the edges and domain policy in testable application modules. TypeScript and Python types are generated from, or checked against, versioned JSON Schema/OpenAPI contracts. A static interface prototype may inform interaction design, but it is not the production system or a source of production data behaviour.

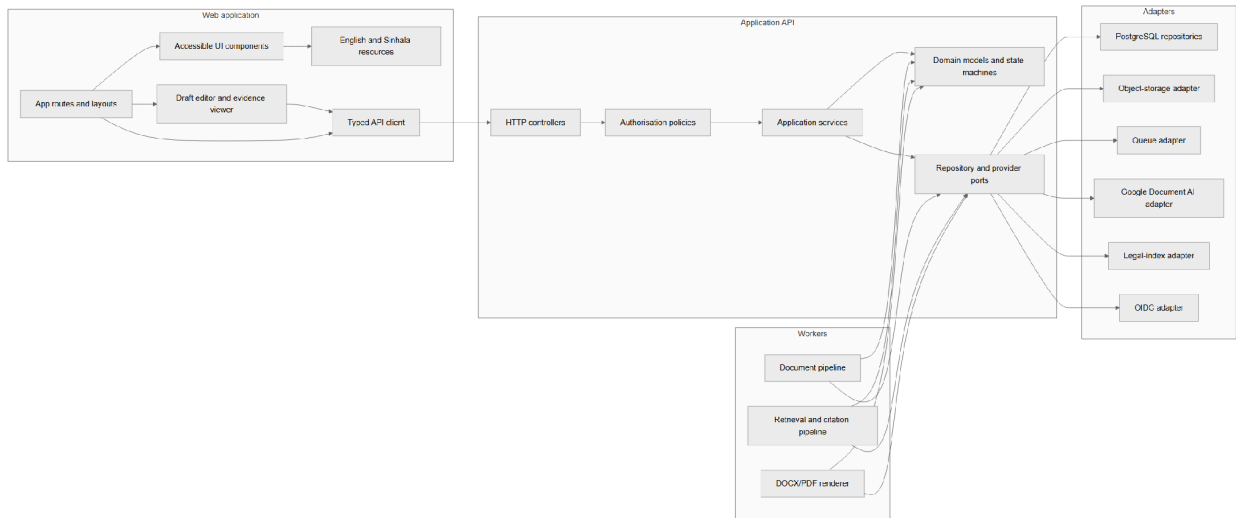


Figure 11. Implementation components and dependency direction.

Figure 11 applies dependency inversion: domain rules do not import cloud SDKs, web controllers, or rendering libraries. Provider and persistence code implements ports owned by the application. This allows deterministic tests to replace Document AI, storage, queues, and search with controlled fixtures.

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8.2 Layers

Layer	Contents	Dependency rule
Presentation	Next.js routes, components, evidence viewer, editor, language resources	Calls the API; never decides legal verification or authorisation
Interface	FastAPI controllers, OpenAPI schemas, command/query mapping	Validates input and delegates to application services
Application	Use-case orchestration, policies, transactions, job submission	Coordinates domain objects and ports
Domain	Matter entities, state machines, invariants, deterministic check contracts	Depends only on stable language libraries and shared schema types
Processing	Document, retrieval, grounding, and export pipelines	Uses application contracts; cannot bypass verification or approval
Infrastructure	PostgreSQL, object storage, queue, identity, Document AI, search, model adapters	Implements ports; provider details remain outside domain packages

8.3 Package Structure

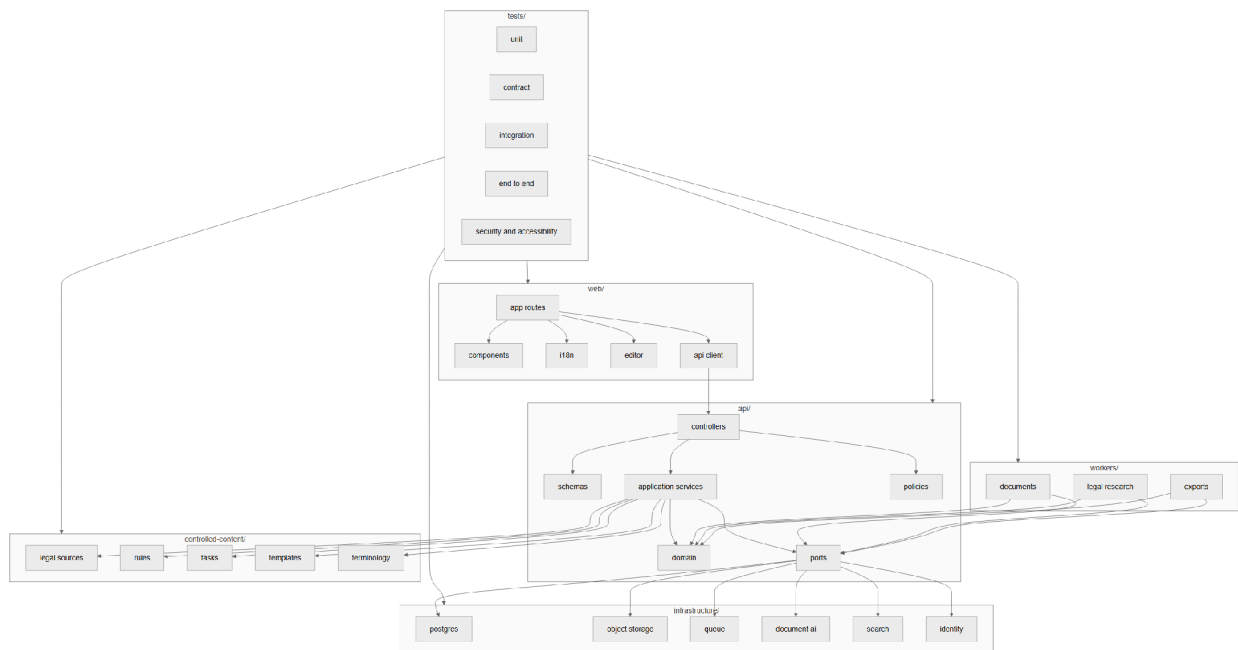


Figure 12. Proposed source-package organisation.

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The package diagram keeps controlled legal content visible as a governed implementation asset rather than burying it in code constants. Tests span contracts and adapters as well as screens. A rule, template, or terminology update follows content governance and does not require an unrelated user interface release.

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8.4 Interfaces and Failure Contracts

Interface	Minimum request/response contract	Failure behaviour
Web API	Versioned JSON over HTTPS; authenticated user; server-derived permissions; correlation ID	Structured safe error; no record existence leak; optimistic concurrency conflict where needed
Upload	Matter ID, file stream, declared metadata; response with document/version/job state	Reject before storage or preserve accepted original and expose recoverable state
Job message	Job ID, record/version ID, operation, schema version, idempotency key	Retry bounded transient failures; permanent failure becomes explicit state
Document processor	Protected content reference or bytes, processor config, expected schema	Timeout/retry/manual fallback; never mutates verified data
Legal search	Query, filters, corpus version; passage IDs, scores, source metadata	Explicit unavailable/insufficient result; no fabricated fallback
Grounded composer	Approved evidence passages and answer schema	Citation validation removes unsupported claims or abstains
Export renderer	Approved draft snapshot, template/version, format, locale	No completed artefact until placeholder, font, checksum, and reopen validation pass
Identity	OIDC code flow and trusted claims	Deny access on failed validation or expired session

All commands that change state carry an expected version or idempotency key. Provider callbacks, if used, are authenticated and matched to a stored job.

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9. Data View

9.1 Data Domains

The system uses four data domains:

1. Confidential evidence: immutable originals, versions, previews, OCR derivatives, exports, and their protected storage metadata.
2. Authoritative matter data: verified/corrected particulars, entities, findings, decisions, drafts, approvals, and audit history.
3. Controlled legal content: statutes, Gazettes, case rules, practice rules, tasks, terminology, templates, versions, approvals, and effective periods.
4. Rebuildable data: search indexes, thumbnails, OCR derivatives, normalised candidate fields, caches, and analytics that can be regenerated without changing authoritative matter data.

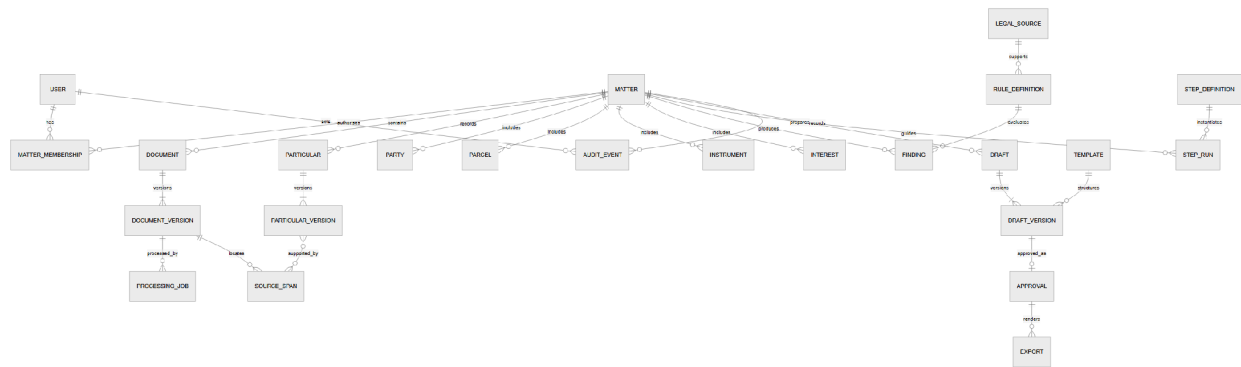


Figure 13. Core data relationships.

Figure 13 shows that authorisation, provenance, legal content, and work products are first-class relationships. Historical versions remain addressable. A replacement document can be excluded from new work without breaking earlier evidence links, approvals, or audit events.

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9.2 State Models

Record	Allowed core states
Document processing	uploaded, queued, extracting, ready-for-review, quality-problem, manual-review, failed, superseded
Particular	unreviewed, verified, corrected, conflict, rejected, blocked, superseded
Finding	pass, warning, fail, needs-review; then open, resolved, accepted, overridden, superseded as applicable
Step run	not-started, ready, blocked, in-progress, completed, overridden, superseded
Draft	preparing, in-review, approved, rejected, superseded
Controlled content	draft, under-review, approved, active, suspended, retired, superseded
Job	queued, running, retryable-failure, failed, completed, cancelled

Transitions are commands checked by domain policy. They record actor, reason, input versions, and event time. A correction adds a new particular version; it does not update the earlier row. An edit to approved content or a draft creates a successor that requires fresh approval.

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9.3 Integrity and Provenance Rules

- An accepted machine-extracted particular has at least one evidence span.
- An evidence span names one immutable document version and a page plus region or character range.
- Only an authorised lawyer can set verified or corrected as a legal verification decision.
- Manual particulars record author, reason, and evidence or missing-evidence status.
- A draft fact binding names the exact verified/corrected particular version.
- An approval names one draft version and content hash.
- An export names one approval, renderer version, format, checksum, and protected object key.
- A finding names the exact rule version, input versions, and authority.
- A grounded claim names resolvable source passage versions.
- Every significant state change creates an audit event in the same database transaction or through a transactional outbox.
- Derived indexes can be rebuilt from controlled sources and approved records.

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9.4 Retention, Backup, and Deletion

Retention is policy-driven by data category and deployment approval. A deletion request first evaluates legal and professional retention duties, active holds, approval history, and backup schedules. The resulting operation removes data no longer authorised while preserving records that must lawfully remain and leaving a non-sensitive audit record of the action.

Backups include the relational database, protected object versions, controlled content, encryption/key recovery material under separate control, and configuration required to resolve evidence links. Restoration tests must show that document-version, particular, decision, approval, export, and audit relationships remain intact.

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10. Size and Performance

10.1 V0 Workload Baseline

The reference workload is a small professional pilot, not an organisation-wide deployment. A typical matter contains up to 15 supported documents, multiple pages per document, hundreds of evidence spans and particulars, tens of findings, and several draft versions. Load tests must publish concurrent-user count, document sizes and pages, network conditions, service quotas, hardware, dataset size, and percentile.

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10.2 Performance Targets and Design Responses

Target	Architecture response
95% of ordinary page and matter operations within 2 seconds	Indexed matter queries, pagination, bounded response schemas, connection pooling, and no OCR/export work in request path
First legal-search page within 3 seconds	Prebuilt versioned index, metadata filters, bounded result count, and cached public-source passages
Grounded answer reaches result, partial result, failure, or abstention within 60 seconds	Separate research job, progress events, strict retrieval/composition time budgets, citation validation, and partial abstention
Upload acceptance acknowledged within 2 seconds	Stream validation, durable original write, job creation, and immediate 202 Accepted response
Up to 15 supported documents reach review or explicit failure within 10 minutes	Parallel jobs within provider quota, page-aware batching, bounded retry, dead-letter/manual review, and per-document progress

Database indexes cover matter membership, matter status, document and version, particular field/state, finding state, draft status, audit time, legal-source metadata, and job status. Large binaries remain in object storage. OCR text and search indexes are partitioned from transactional reads.

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10.3 Capacity Evolution

Metrics gathered in V0 include queue delay, processing duration by class/page, provider error and retry counts, search latency, draft/export duration, database query percentiles, storage growth, concurrent users, and manual-review rate. V1 to V3 capacity decisions use these measurements. Services may be split or independently scaled only when measured load or isolation needs justify the operational cost.

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11. Quality

11.1 Quality Attribute Scenarios

Attribute	Scenario and required response
Legal safety	If a mandatory field is unverified or a citation does not resolve, approval or the unsupported claim is blocked
Security	A user requests another matter by guessed ID; server policy denies it without confirming the record exists and logs a safe event
Privacy	A processing job fails; logs contain job/provider codes but no document page, identity number, signature, or full address
Reliability	Document AI times out; the original remains intact and the job retries or reaches explicit manual review
Auditability	A lawyer corrects a parcel number and later approves a draft; history identifies old/new values, evidence, actor, rule results, draft hash, and approval
Maintainability	A Gazette changes a form; an administrator adds and approves a new effective-dated template/rule version without rewriting historical matters
Accessibility	A keyboard and screen-reader user completes upload, evidence review, correction, approval, and export at WCAG 2.2 Level AA
Bilingual fidelity	Sinhala and English source text, entered particulars, and approved output render without script loss in the browser and reopened exports
Recoverability	A database and object-store restore reconstructs the exact evidence-to-fact-to-draft-to-approval links
Performance	A 15-document matter processes asynchronously while ordinary matter work remains responsive

11.2 Verification Strategy

Level	Focus
Unit	State transitions, normalisation, form bindings, deterministic checks, citation rules, permission predicates
Contract	OpenAPI/JSON Schema compatibility; worker messages; provider-normalisation fixtures; export manifests
Integration	PostgreSQL transactions, object references,

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Level	Focus
	queue retry/idempotency, Document AI adapter, legal index, renderer
End-to-end	Create matter, upload, manual/automatic processing, verify, check, research, draft, approve, export, inspect history
Security	Cross-matter isolation, role escalation, upload attacks, signed-link expiry, secrets/log review, approval bypass, common web risks
Legal/domain	Lawyer-labelled extraction, evidence, rule, finding, citation, wording, and draft correctness with abstentions reported
Accessibility	Automated checks plus keyboard, zoom, focus, screen-reader, English/Sinhala, and print/export review
Recovery	Backup restore, failed-job recovery, provider outage, index rebuild, export regeneration, and audit reconciliation

Release-blocking failures include an invented citation, unsupported retained legal claim, cross-matter disclosure, silent loss of an accepted original, mandatory unverified fact in an approved export, altered locked wording, unresolved hidden placeholder, or approval attached to changed content.

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11.3 Observability

Each request and background job carries a correlation ID. Metrics report counts and timings by privacy-safe class and state. Alerts cover failed or stalled jobs, provider errors, storage/database health, unusual authorisation denials, audit-write failure, search-index version mismatch, and export validation failure. Logs use stable record IDs and error codes instead of client content.

11.4 Known Limitations and Planned Hardening

- V0 does not claim reliable extraction of handwriting, severely degraded pages, or unsupported scripts. These produce a visible manual-review path.
- Google Document AI language, processor, version, and regional capability must be verified against the approved V0 document set. Interface and export support for Sinhala does not imply reliable Sinhala source OCR.
- Abstract of Title forms and historical chain reconstruction are outside V0.
- Form 8 transfer is the first active prescribed instrument. Other forms remain inactive until their schemas, wording, checks, and tests receive domain approval.
- Live hosting, retention, deletion, external processing, professional conduct, and numerical evaluation thresholds require formal approval.

These limitations are represented as product states and test cases rather than hidden assumptions. Later releases may improve hard-document processing and widen notarial coverage after separate evaluation.

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